

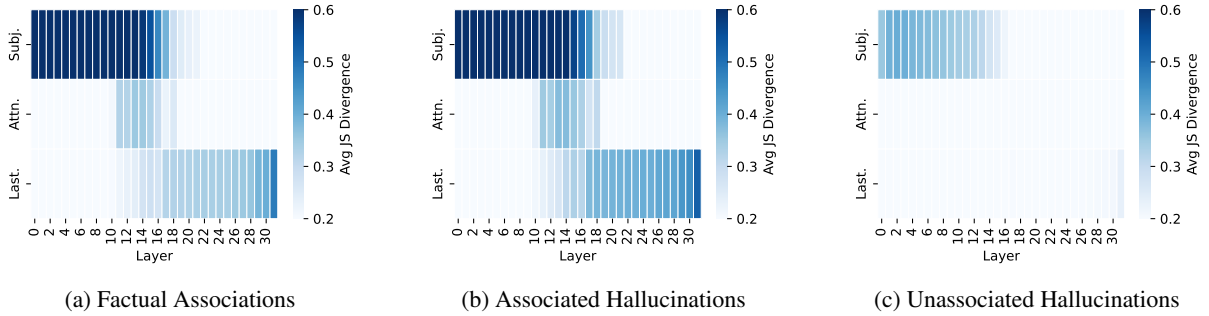


Figure 2: Effect of interventions across layers of LLaMA-3-8B. The heatmap shows JS divergence between the output distribution before and after intervention. Darker color indicates that the intervened hidden states are more causally influential on the model’s predictions. Top row: patching representations of subject tokens. Middle row: blocking attention flow from subject to the last token. Bottom row: patching representations of the last token.

	LLaMA-3-8B	Mistral-7B-v0.3
Factual Association	3,506	3,354
Associated Hallucination	1,406	1,284
Unassociated Hallucination	7,381	7,655
Total	12,293	12,293

Table 1: Dataset statistics across categories.

tion. Given a factual query prompt (e.g., “*Barack Obama was born in the city of*”), a model typically follows three steps to produce the correct completion (e.g., “*Honolulu*”). First, early layers of LLMs construct enriched subject representations encoding attributes specific to the subject (e.g., “*Barack Obama*”). Second, attention modules in middle layers propagate this subject information to the prediction position (i.e., the last token). Finally, upper layers extract the relevant attribute (e.g., “*Honolulu*”) from the propagated information to produce the next token.

While this mechanism has been established for factual generation, the internal processes underlying hallucinations remain less understood. We hypothesize that hidden states primarily reflect *how* the model leverages its parametric knowledge, rather than whether the output is factually correct. Consequently, hallucinations produced through statistical shortcuts or spurious correlations may follow internal processes similar to factual recall.

To test this hypothesis, we categorize hallucinatory outputs based on their reliance on learned associations with the subject entity. Specifically, we perform a causal intervention by blocking attention from the subject tokens and measuring the resulting shift in the output distribution. Because attention is the primary mechanism through which subject representations interact with subsequent tokens, a large distributional shift under masking

indicates a strong reliance on the subject representation. We quantify this shift using the Jensen–Shannon (JS) divergence between the original and masked output distributions. We set the threshold to the average JS divergence observed for all correct answers (FAs): hallucinations are categorized as **Associated Hallucinations (AH)** if the JS divergence exceeds this threshold; otherwise, they are categorized as **Unassociated Hallucinations (UH)**. Table 1 summarizes the final data statistics.

4 Analysis of Internal States in LLMs

As established by Geva et al. (2023), factual generation relies on three components: (i) the enrichment of subject representations in the early layers, (ii) an attention flow from the subject to the last token in the middle layers, and (iii) the decoding of the last token representation in the upper layers. In this section, we first visualize this overall information flow across all categories (§4.1), and analyze each component in §4.2 through §4.4. Parallel experimental results on Mistral are summarized in Appendix C.

4.1 Visualization of Information Flow

In §3.3, we categorized hallucinations by intervening on the attention flow from the subject tokens. To provide a more granular understanding of the internal mechanisms established by Geva et al. (2023), we perform fine-grained causal interventions on all three components across each knowledge category. This allows us to examine whether the enrichment and decoding phases remain consistent between Associated Hallucinations (AHs) and Factual Associations (FAs).

Experimental Setup We measure the importance of each component by observing the resulting shift in the output distribution after intervention; a

large shift indicates the component is critical for the prediction, while a small shift suggests a limited role. We quantify each shift using Jensen-Shannon (JS) divergence between the original and intervened output distributions. Specifically, for subject-token and last-token representations, we corrupt the corresponding hidden states at each layer ℓ . For the attention mechanism, we follow Geva et al. (2023) to mask the attention flow between the subject and last tokens at layer ℓ , using a window size of 5 layers.

Experimental Results As shown in Figure 2, FAs and AHs share highly similar information flows. The components that are crucial for producing factually correct outputs are also crucial for producing AHs. This pattern suggests that, despite being factually incorrect, AHs are generated through the same underlying internal mechanisms as factual recall. In contrast, UHs display substantially weaker information flow, reflecting their limited reliance on the subject entity during generation.

These findings indicate that hidden states primarily capture **whether the model is recalling parametric knowledge**, rather than **whether the generated output is factually correct**. These results challenge the prevailing assumption in prior works (Azaria and Mitchell, 2023; Gottesman and Geva, 2024; Yüsekçönül et al., 2024; Orgad et al., 2025; Varshney et al., 2023b) that internal representations inherently encode factual correctness. Instead, our results show that factual outputs and association-driven hallucinations can share nearly identical internal computational pathways.

It is worth noting that the strong causal effect of attention-flow interventions for FAs and AHs is expected, since our labeling procedure already uses attention masking to distinguish AHs from UHs. However, interventions on subject-token representations and last-token hidden states provide an independent validation of our hypothesis. These components remain highly influential for both FAs and AHs, confirming that the shared mechanism extends beyond the attention pathways used during labeling.

4.2 Analysis of Subject Representations

Having established the overall information flow, we next analyze the subject representations in the early layers. Our goal is to understand why the internal behavior at subject positions is highly similar for

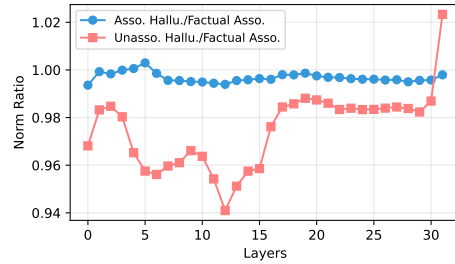


Figure 3: Norm ratio curves of subject representations in LLaMA-3-8B, comparing AHs and UHs against FAs as the baseline.

AHs and FAs, yet differs substantially for UHs. To this end, we first examine the norm of these representations (§4.2.1), and then analyze how this behavior correlates with subject prevalence in the pre-training data (§4.2.2).

4.2.1 Norm of Subject Representations

We measure the average L_2 norm of subject representations for each knowledge category across layers (details in Appendix B). To facilitate direct comparison, we report the L_2 norms of both hallucination types (AHs and UHs) as ratios relative to the norm of factually correct predictions (FAs). A ratio close to 1 indicates norms comparable to FAs, while ratios below 1 indicate smaller norms.

As shown in Figure 3, for LLaMA-3-8B, the norm of AH samples is highly comparable to that of FAs, with ratios remaining close to 1 across layers. In contrast, UH samples consistently exhibit smaller norms than FAs in nearly all layers, with the ratio dropping to approximately 0.95 in the middle layers (layers 10–15). This suggests that **AHs and FAs activate subject representations with similar strength, whereas UHs involve weaker subject encoding**. We further analyze the subspace overlap between subject hidden states and MLP weight matrices in Appendix D.1, which provides additional evidence that AHs and FAs occupy highly similar representational regions.

4.2.2 Correlation with Subject Popularity

To explain why AHs exhibit subject representation norms comparable to FAs while UHs remain weaker, we hypothesize that this difference arises from subject popularity in the pre-training data. Prior work (Kang et al., 2024) shows that entities appearing more frequently during pre-training tend to produce hidden representations with larger norms. Accordingly, we expect that FAs and AHs predominantly involve high-popularity entities,

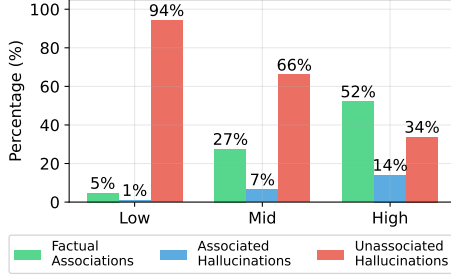


Figure 4: Sample distribution across different subject popularity (low, mid, high) in LLaMA-3-8B, measured by monthly Wikipedia page views.

while UHs are more common for low-popularity entities.

Following [Mallen et al. \(2023a\)](#), we use average monthly Wikipedia page views as a proxy for subject frequency during pre-training. We partition subjects into three popularity bins and analyze the distribution of FAs, AHs, and UHs across them. The distribution in Figure 4 confirms that the prevalence of both FAs and AHs is strongly associated with subject popularity. In the lowest popularity bin, both FAs and AHs are rarely observed (5% for FAs and 1% for AHs). In contrast, their prevalence significantly increases in the highest popularity bin (52% for FAs and 14% for AHs). However, the prevalence of UHs shows an inverse trend: while the lowest popularity bin is dominated by UHs (94%), their prevalence drops to 34% in the highest popularity bin.

This observation highlights a critical implication: if high-popularity subjects account for a substantial portion of real-world language model usage, AHs may constitute a substantial portion of hallucinations encountered in practical deployments. Meanwhile, as AHs arise mainly on popular subjects, they are often indistinguishable from FAs by popularity-based heuristics, contradicting prior work ([Mallen et al., 2023a](#)) that links popularity to hallucinations.

4.3 Visualization of Attention Flow

Having examined how the model forms subject representations, we next visualize how this information is propagated to the last token of the input where the model generates the object of a knowledge tuple. While the strong reliance of AHs on this attention flow is an expected consequence of our causal intervention labeling, visualizing this mechanism explicitly illustrates the divergence between AHs/FAs and UHs, providing a more com-

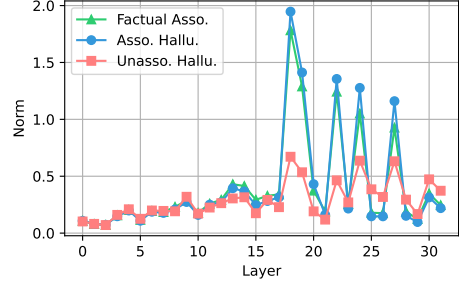


Figure 5: Subject-to-last attention contribution norms across layers in LLaMA-3-8B. Values show the norm of the attention contribution from subject tokens to the last token at each layer.

prehensive view of the internal mechanism behind how the model produces outputs for each category.

To quantify the specific contribution from subject tokens ($s_1, \dots, s_n$) to the last token, we compute the attention contribution from subject tokens to the last position:

$$\mathbf{a}_{\text{last}}^\ell = \sum_k \sum_h A_{\text{last}, s_k}^{\ell, h} (\mathbf{h}_{s_k}^{\ell-1} W_V^{\ell, h}) W_O^{\ell, h}. \quad (1)$$

where $A_{i,j}^{\ell, h}$ denotes the attention weight assigned by the h -th head in the layer ℓ from the last position i to subject token j . Here, $\mathbf{a}_{\text{last}}^\ell$ represents the subject-to-last attention contribution at layer ℓ . Intuitively, if subject information is critical for prediction, this contribution should have a large norm; otherwise, the norm should be small.

Figure 5 shows that in LLaMA-3-8B, both AHs and FAs exhibit large attention-contribution norms in mid-layers, indicating a strong information flow from subject tokens to the target token. In contrast, UHs show consistently lower norms, implying that their predictions rely far less on subject information. [Yüksekgönül et al. \(2024\)](#) previously argued that high attention flow from subject tokens signals factuality and proposed using attention-based hidden states to detect hallucinations. Our results challenge this view: **the model propagates subject information just as strongly when generating AHs as when producing correct facts.**

4.4 Analysis of Last Token Representations

Since both FAs and AHs involve strong information propagation from subject representations, whereas this propagation is weaker for UHs, we hypothesize that this disparity in information flow also leads to distinct geometric properties in the last token representations. Specifically, the strong subject propagation for FAs and AHs suggests that their last-token representations receive significant subject-specific updates, causing them to *diverge* in